

EX N0:09
30.09.26

FORWARD CHAINING

Exercise 1: Simple Inference

```
:- dynamic known/1.  
% Initial knowledge  
start :- assert(known(a)), assert(known(b)), infer.  
% Rule  
infer :- known(a), known(b), \+ known(c), assert(known(c)), write(&#39;Inferred  
c&#39;), nl.  
% Query  
% ?- start.
```

Output:

Inferred c

Exercise 2: Fire Alert System

```
:- dynamic known/1.  
% Rule to infer fire  
check :- known(smoke), known(heat), assert(known(fire)), write(&#39;Fire  
Alert!&#39;), nl.  
% Start rule  
start :- assert(known(smoke)), assert(known(heat)), check.  
% Query  
% ?- start
```

Output:

Fire Alert!

Exercise 3: Water Purity Detection

```
:- dynamic known/1.  
% Rule to infer pollution  
infer_pollution :- known(cloudy_water), known(bad_smell), assert(known(polluted)),  
write(&#39;Water is polluted&#39;), nl.  
start :- assert(known(cloudy_water)), assert(known(bad_smell)), infer_pollution.  
% Query  
% ?- start
```

Output:

Water is polluted

Result:

Hence, the process of performing the forward chaining operations is verified successfully.